



**Bachelor of Information Technology**

**School of Innovation, Design and Technology**

**SD7501 – Web Application Development**

**ASSIGNMENT 2 – Web Application Development Project (Phase 1)**

**Trimester 2, 2026**

**Assessment weighting:** 35%

**Due date and time:** Friday, 18<sup>th</sup> of September 2026, 23:59

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## Assessment Overview

This assessment requires students to work in teams to design, develop, document, and present a secure web application using ASP.NET Core MVC and related technologies.

The project provides learners with the opportunity to apply the concepts and techniques covered in the course by developing a realistic web-based solution to an approved business or community problem. Students will demonstrate their ability to design software architecture, implement database-driven functionality, apply software engineering principles, and document the development process.

## Course Aim

To enable the learner to:

- Evaluate and apply the use of appropriate platform and architecture, for the development of web applications.
- Integrate applications with a database and learn how to access web data using managed data providers and objects.
- Investigate the security challenges and security models for web applications.

## Learning Outcomes Assessed

This assessment is designed to assess the following learning outcomes. On successful completion of this assessment, you will be able to:

LO3: Design and implement an appropriate secure internet application solution to an unstructured problem.

## Due Date, Format and Submission Instructions

This assessment is due by **Friday, 18<sup>th</sup> of September 2026, 23:59**.

Although this is a group project, **each student is required to submit their own copy of the assessment**.

You must submit **a single ZIP file** via the submission link provided in Moodle. The ZIP file must contain:

- The complete application project folder with all source code files.
- The project documentation, including all required contents specified in Task 2.

## Pass Requirements

To pass a course a learner must:

- Attempt all assessments
- Achieve an average mark of 50% or above over all assessments,
- Achieve all learning outcomes

## Assessment Task and Requirements

### Task Description

You are required to work as a development team of **two to three members** to design and implement a web application using ASP.NET Core MVC.

**The project topic must be approved by your tutor before development begins.** The same project will continue into the Final Project assessment. You are expected to demonstrate your understanding of web development concepts, software engineering principles, database design considerations, and ASP.NET Core development practices through the planning, design, and initial development of the proposed application. Assignment 2 should establish a strong foundation for the Final Project, where the core features, advanced functionality, and full implementation of the application will be completed and refined.

The final solution should be an original application developed by the team and must not rely on scaffolding-generated solutions, templates, or copied source code.

### Content Requirements

#### Task 1: Application Design and Implementation (65 marks)

You are required to work as a **team to design and develop** an ASP.NET Core MVC web application that demonstrates the core concepts covered in the class.

Your application must include:

- a. ASP.NET Core MVC (Model-View-Controller)
- b. Entity Framework Core 6 - SQL Server
- c. Dependency Injection
- d. Repository Pattern and UnitOfWork implementation
- e. *CRUD operations* on the Model class(es).
- f. Accessing posted data using *model binding*.
- g. Server-side / *Client-Side Validations using JQuery*.
- h. Routing and Navigation of your application should be configured correctly and should **NOT** lead to an empty page.
- i. Quality of User Interface (UI) using Bootstrap

#### Task 2: Project Documentation (35 marks)

Prepare an **individual** project report that includes the following contents:

- a. Provide a clear description of the application being developed, including its purpose, key features, project deliverables, and both functional and non-functional requirements.

- b. Present a conceptual framework that outlines the overall development plan and direction for the project. This should illustrate how the application will evolve throughout both Assignment 2 and the Final Project, including the proposed architecture, major components, and development milestones.
- c. Document the key classes, methods, and key features implemented in the application. Explain their purpose and how they contribute to the overall functionality and design of the system.
- d. Discuss the challenges and constraints encountered during the development process and explain the strategies, techniques, and solutions employed to address them. This section should demonstrate reflection on the development experience and decision-making process.
- e. Include the **GitHub repository link** containing all project resources.

## Academic Requirements

### Extensions and Late Submission

If you are unable to submit an assessment by the due date due to circumstances such as illness or unforeseen events, you may apply for an extension. You must submit an extension request to the course lecturer and provide appropriate supporting evidence. Requests should be made as early as possible and **before the due date**.

Late submissions without an approved extension will result in a mark of zero, in line with programme and institutional policy. Submission of the wrong assignment will also result in a mark of zero. Further information about extensions, late submission, and further assessment opportunities is available in the Programme Handbook.

## MARKING SCHEDULE

This assessment is marked out of 100 marks and contributes 35% to the final course grade.

| Criterion                                                   | What is Assessed                                                                                                 | Learning Outcomes Assessed | Marks Available | Marks Awarded |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------|---------------|
| <b>Task 1 - Application Design and Implementation – 65%</b> |                                                                                                                  |                            |                 |               |
| Accuracy and Completeness                                   | Degree to which the application satisfies the specified requirements and functions without runtime errors.       | LO3                        | 10              |               |
| User Interface and Navigation                               | Effective use of Bootstrap, consistency of design, usability, routing, and navigation.                           | LO3                        | 10              |               |
| Overall Quality and Functionality                           | Application robustness, reliability, and overall quality of implementation.                                      | LO3                        | 10              |               |
| CRUD Operations                                             | Successful implementation of CRUD functionality for at least two model classes.                                  | LO3                        | 15              |               |
| Repository Pattern and Unit of Work                         | Correct implementation of repository and Unit of Work patterns.                                                  | LO3                        | 5               |               |
| Validation                                                  | Implementation of both server-side and client-side validation.                                                   | LO3                        | 10              |               |
| GitHub Contribution                                         | Evidence of meaningful contributions from all team members through GitHub commits and version control practices. | LO3                        | 5               |               |
| <b>Task 2 – Project Documentation – 35%</b>                 |                                                                                                                  |                            |                 |               |
| Application Description and Requirements Specification      | Clarity and completeness of the application overview, requirements, and deliverables.                            | LO3                        | 5               |               |
| Conceptual Framework                                        | Quality of the conceptual framework, architecture, development plan, and project roadmap.                        | LO3                        | 5               |               |

|                                        |                                                                                |     |            |  |
|----------------------------------------|--------------------------------------------------------------------------------|-----|------------|--|
| Classes, Methods, and Key Features     | Documentation and explanation of classes, methods, and implemented features.   | LO3 | 10         |  |
| Constraints and Development Strategies | Discussion of challenges encountered and strategies used during development.   | LO3 | 10         |  |
| Written communication and Referencing  | Structure, formatting, professional presentation, and appropriate referencing. | LO3 | 5          |  |
| <b>TOTAL</b>                           |                                                                                |     | <b>100</b> |  |

## MARKING RUBRIC

| Criteria                                                    | Excellent<br>(80% – 100%)                                                                           | Competent<br>(60% - 79%)                                             | Basic<br>(40% - 59%)                                                   | Limited<br>(0% - 39%)                                                    |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Task 1 - Application Design and Implementation – 65%</b> |                                                                                                     |                                                                      |                                                                        |                                                                          |
| Accuracy and Completeness<br>(10 marks)                     | Meets all requirements and operates without errors.<br>(8 – 10 marks)                               | Meets most requirements with minor issues.<br>(6 – 7 marks)          | Meets basic requirements; some functionality missing.<br>(4 – 5 marks) | Major requirements missing or frequent errors.<br>(0 – 3 marks)          |
| User Interface and Navigation<br>(10 marks)                 | Professional, consistent design with effective Bootstrap and seamless navigation.<br>(8 – 10 marks) | Good design with minor UI or navigation issues.<br>(6 – 7 marks)     | Basic interface and navigation.<br>(4 – 5 marks)                       | Poor design, inconsistent layout, or broken navigation.<br>(0 – 3 marks) |
| Overall Quality and Functionality<br>(10 marks)             | Robust, reliable, and well-implemented application.<br>(8 – 10 marks)                               | Functional with minor issues.<br>(6 – 7 marks)                       | Adequate functionality with some weaknesses.<br>(4 – 5 marks)          | Limited functionality or unreliable application.<br>(0 – 3 marks)        |
| CRUD Operations<br>(15 marks)                               | Complete and correct CRUD implementation for at least two model classes.<br>(12 – 15 marks)         | Most CRUD functions implemented with minor issues.<br>(9 – 11 marks) | Partial CRUD implementation.<br>(6 – 8 marks)                          | CRUD functionality largely incomplete or missing.<br>(0 – 5 marks)       |
| Repository Pattern and Unit of Work<br>(5 marks)            | Correct and well-structured implementation.<br>(4 – 5 marks)                                        | Mostly correct implementation.<br>(3 marks)                          | Basic implementation with some issues.<br>(2 marks)                    | Incomplete or not implemented.<br>(0 – 1 mark)                           |
| Validation<br>(10 marks)                                    | Comprehensive server-side and client-side validation.<br>(8 – 10 marks)                             | Good validation with minor omissions.<br>(6 – 7 marks)               | Basic validation implemented.<br>(4 – 5 marks)                         | Limited or no validation.<br>(0 – 3 marks)                               |
| GitHub Contribution<br>(5 marks)                            | Clear and meaningful contributions from all team members.<br>(4 – 5 marks)                          | Good evidence of team contributions.<br>(3 marks)                    | Limited evidence of collaborative work.<br>(2 marks)                   | Minimal or no contribution evidence.<br>(0 – 1 mark)                     |

| Criteria                                                         | Excellent<br>(80% – 100%)                                                                        | Competent<br>(60% - 79%)                              | Basic<br>(40% - 59%)                                    | Limited<br>(0% - 39%)                                    |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| <b>Task 2 – Project Documentation – 35%</b>                      |                                                                                                  |                                                       |                                                         |                                                          |
| Application Description and Requirements Specification (5 marks) | Clear, detailed, and complete description and requirements. (4 – 5 marks)                        | Mostly clear with minor omissions. (3 marks)          | Basic description provided. (2 marks)                   | Incomplete or unclear. (0 -1 mark)                       |
| Conceptual Framework (5 marks)                                   | Well-developed framework, architecture, and project roadmap. (4 – 5 marks)                       | Good framework with reasonable detail. (3 marks)      | Basic framework presented. (2 marks)                    | Weak or missing framework. (0 – 1 mark)                  |
| Classes, Methods, and Key Features (10 marks)                    | Comprehensive documentation and explanation of classes, methods and key features. (8 – 10 marks) | Good documentation with minor gaps. (6 – 7 marks)     | Basic documentation provided. (4 – 5 marks)             | Limited or missing documentation. (0 – 3 marks)          |
| Constraints and Development Strategies (10 marks)                | Insightful discussion of challenges and solutions. (8 – 10 marks)                                | Good discussion with relevant examples. (6 – 7 marks) | Basic discussion with limited reflection. (4 – 5 marks) | Minimal discussion or reflection. (0 – 3 marks)          |
| Written communication and Referencing (5 marks)                  | Professional presentation, excellent structure, and correct referencing. (4 – 5 marks)           | Good presentation with minor issues. (3 marks)        | Acceptable presentation and referencing. (2 marks)      | Poor structure, formatting, or referencing. (0 – 1 mark) |